

Numerical MCQs on Impurity Metrics and Tree Splitting

Question 1: Consider a node with two classes with probabilities 0.7 and 0.3. What is the Gini impurity for this node?

A. 0.42

B. 0.21

C. 0.49

D. 0.51

Answer: A. 0.42

Question 2: For a node with class probabilities 0.8 and 0.2, what is the entropy of the node? (Use log base 2)

A. 0.50

B. 0.7219

C. 0.92

D. 1.0

Answer: B. 0.7219

Question 3: A node has a 200 data points labeled '1' and 300 data points labeled '0'. What is the misclassification error for this node?

A. 0.4

B. 0.3

C. 0.6

D. 0.2

Answer: A. 0.4

Question 4: A parent node has a Gini impurity of 0.5. It is split into two child nodes with Gini impurities of 0.2 and 0.3, and the child nodes contain 40% and 60% of the data, respectively. What is the reduction in impurity (weighted reduction) due to this split?

A. 0.24

B. 0.26

C. 0.18

D. 0.20

Answer: A. 0.24

Question 5: A parent node has a misclassification error of 0.35. After a split, the two child nodes have misclassification errors of 0.1 and 0.2, containing 30% and 70% of the samples respectively. What is the overall impurity reduction from this split?

- A. 0.18
- B. 0.20
- C. 0.22
- D. 0.15

Answer: A. 0.18

Multiple Choice Questions on Decision Trees

Question 1: Which of the following is not a key part of the structure of a tree?

- A. Root
- B. Parent
- C. Sibling
- D. Leaf

Answer: C. Sibling

Question 2: What is the primary purpose of using decision trees in machine learning?

- A. To partition the feature space into regions with similar labels
- B. To calculate distances between data points
- C. To cluster data randomly
- D. To perform gradient descent

Answer: A. To partition the feature space into regions with similar labels

Question 3: Which impurity metric is defined as 1 minus the maximum proportion of a class in a region?

- A. Gini Impurity
- B. Entropy
- C. Squared loss
- D. Misclassification error

Answer: D. Misclassification error

Question 4: What does the Gini Impurity Index measure in a decision tree?

- A. The amount of information gain
- B. The homogeneity of a node
- C. The distance between features
- D. The variance of a feature

Answer: B. The homogeneity of a node

Question 5: In node splitting, what two elements are primarily determined?

- A. The feature to split on and the loss function
- B. The threshold and the feature to split on
- C. The child nodes and the parent node
- D. The depth and the height

Answer: B. The threshold and the feature to split on

Question 6: Which method is used to prevent overfitting in decision trees by reducing the size of a tree with a large depth?

- A. Increasing `max_depth`
- B. Cost-complexity pruning
- C. Reducing `min_samples_leaf`
- D. Removing the root node

Answer: B. Cost-complexity pruning

Question 7: What Scikit-Learn parameter specifies the minimum number of samples required to form a leaf node in a decision tree?

- A. `min_samples_split`
- B. `min_samples_leaf`
- C. `max_depth`
- D. `cost_complexity`

Answer: B. `min_samples_leaf`

Question 8: In ID3, which condition(s) below result(s) in a node becoming a leaf node?

- A. When there are no more features on which to split.
- B. When further splitting does not reduce impurity

- C. When all data points in the node have the same label
- D. When the node has the highest depth

Answer: A. When there are no more features on which to split. AND C. When all data points in the node have the same label

Question 9: Which impurity metric uses the expression: $\sum_c \hat{p}_c(1 - \hat{p}_c)$?

- A. Entropy
- B. Gini Impurity
- C. Loss function
- D. Variance

Answer: B. Gini Impurity

Question 10: What is the underlying assumption of decision trees about the distribution of data points?

- A. Data points are more or less randomly distributed
- B. Data points form more or less homogeneous clusters
- C. Data points have equal probability across all regions
- D. Data points follow a uniform distribution

Answer: B. Data points form homogeneous clusters

Question 11: In regression trees, how is the predicted label for a region calculated?

- A. As the mean of the target values in that region
- B. As the mode of the target values
- C. As the median of the target values
- D. As the maximum of the target values

Answer: A. As the mean of the target values in that region

Question 12: Which parameter in decision trees limits the maximum depth of the tree to control overfitting?

- A. `min_samples_split`
- B. `max_depth`
- C. `min_samples_leaf`
- D. `cost_complexity`

Answer: B. `max_depth`

Question 13: The function implemented by a decision tree uses regions corresponding to leaves. What does each leaf represent?

- A. A split threshold
- B. A decision rule
- C. A region with a uniform label
- D. A feature index

Answer: C. A region with a uniform label

Question 14: As the regularization parameter in $R_\alpha(T) = R(T) + \alpha|T|$ increases,

- A. the risk of overfitting reduces
- B. the risk of underfitting reduces
- C. model variance tends to increase
- D. model bias tends to increase

Answer: A. the risk of overfitting reduces, and D. model bias tends to increase.

Question 15: What does cost-complexity pruning aim to balance in a decision tree model?

- A. The trade-off between training error and model interpretability
- B. The trade-off between misclassification error and the number of leaves
- C. The trade-off between depth and width
- D. The trade-off between regression and classification accuracy

Answer: B. The trade-off between misclassification error and the number of leaves